Brandon Ho

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Experience

Relativity Space Long Beach, CA
Robotics Software Engineer II *September 2024 - Current*

- Building a flexible, **real-time** robotics software platform (**C++**) to support large metal additive manufacturing
- **ROS Middleware**: Built novel RT-capable **Zenoh+EVL**-based RMW for in-band/out-of-band comms, increasing throughput **10x**, reliability **1.5x**, reducing codebase redundancy **15%**
- **Real-Time Safety & Determinism**: Built safety-critical RT monitoring system tracking loop overruns and packet drops; implemented **dynamic TDMA throttling** to prevent cycle overruns by predicting write times and deferring to next cycle; designed middleware packet reordering and stale packet dropping for data fidelity; profiled RT processes to optimize worst-case timing; validated on **HITL** systems with comprehensive test coverage
- **Robot-Sensor Coordination**: Enabled continuous 3D scanning by integrating a **Gocator 2D laser profilometer** on 6-DOF robot; implemented real-time **RANSAC circle fitting** at **10Hz** and **ICP registration** to compensate for robot positioning inaccuracies, increasing data completeness **20x** and reducing noise by **50%**
- **Closed-loop Control**: Implemented **PID** w/ anti-windup for wire feed speed, **reducing bead variation by 30%**
- **RT Linux Kernel**: Automated **Xenomai4** build/packaging via **Ansible**; contributed upstream patches fixing network stack packet leak and page faults, **improving system safety during robot operation (90% crash reduction)**
- **Project Management**: Successfully executed **1-year R&D project** to revamp core SW middleware with team of 3 engineers, wrote roadmap, presented monthly to stakeholders, delegated work via **Jira**

Relativity Space Long Beach, CA
Robotics Software Engineer Intern *June 2023 - September 2023*

- Exposing **OPC-UA** interface on Fronius welders to real-time context for Wire Arc Additive Manufacturing
- **ROS2 Driver**: Wrote an OPC-UA driver in **ROS2 (C++)** to read and write over RT (shared memory) and non-RT contexts, exposing **40+ new parameters** and increasing controllable welding parameters by **5x**
- **System Design**: Refactored welder state machine and inverted object dependencies, **reducing 1000+ lines of code**

InflammaSense La Jolla, CA
Software Developer *January 2021 - September 2022*

- Built medical device to record and analyze neural signals for early signs of sepsis, published **Scientific Reports** (link)
- **Data Pipeline**: Designed a robust async data pipeline with the **ZeroMQ** Python API for the wireless display, transmission, storage (**PostgreSQL**), analysis of neural data for an array of **100+ medical devices (~155 Mbps)**
- **Project Management**: Documented a comprehensive system model and **leading a team of 5+ SWE** with Asana
- **UI Dev**: Built a GUI to display high throughput EKG data (8 kHz) w/ **PyQt**, enabling real-time data monitoring

Education

University of California San Diego GPA: 3.93/4 (M.S.)
B.S. Computer Engineering (Magna Cum Laude), M.S. Intelligent Systems, Robotics & Control *June 2024*

- **Relevant Coursework**: Software Engineering, Deep Learning, Operating Systems, Computer Vision, Autonomous Vehicles, Sensing & Estimation, Planning & Learning, Robot Sys Design & Implementation, Statistical Learning

Projects

Autonomous Rover Path Planning Yonder Dynamics
Software Lead *September 2020 - June 2023*

- Lead software dev for rover to autonomously path and perform tasks in the University Rover Competition
- **Autonomous Exploration**: Implemented RRT* search in C++, **increasing rover exploration speed ~10x**
- **Obstacle Avoidance**: Created a **ROS** obstacle detection and avoidance pipeline in C++ with **RealSense** stereo camera to efficiently convert 3D pointclouds at 30Hz into occupancy grid and output avoidance waypoints

Skills

Programming Languages: Python, C++, Java, SQL, JS, PHP, HTML/CSS, TypeScript
Technologies: AWS, Git, Jupyter Notebook, Linux, Docker, Confluence, Jira, Gerrit, Gitlab CI, Claude
Frameworks: ROS2, OPC-UA, PyTorch, MQTT, Zenoh, Xenomai, NumPy, Pandas, SciKit, Ansible